

Semester Project (2023)

**Ramsey, size-Ramsey and induced size-Ramsey
numbers**

Tanish Patil

Student:	Tanish Patil
Thesis Supervisor:	Prof. Dr. Benny Sudakov
Co-supervisors:	Domagoj Bradač Dr. Nemanja Draganić
Date:	August 31, 2023

Contents

1	Ramsey numbers	1
1.1	Diagonal Ramsey numbers	1
1.2	Off-diagonal Ramsey numbers	5
1.3	Multicoloured Ramsey numbers and paths, cycles and trees	9
1.4	Conjectures	13
2	Size-Ramsey numbers	14
2.1	Complete graphs	14
2.2	Paths and bounded-degree trees	17
2.3	More general graphs	22
2.4	Conjectures	23

3	Induced Ramsey numbers	24
3.1	Cycles and paths	24
3.2	Trees and bounded-degree graphs	28
3.3	Conjectures	30

Abstract

This paper is a survey of notable results in graph theory concerning the size-Ramsey and induced size-Ramsey numbers of graphs. It seeks to discuss existing bounds as well as summarising some of the methods and proofs used to obtain these.

Acknowledgements. Thank you to Professor Benny Sudakov for acknowledging to supervise my semester project, and a very special thanks to his PhD students Domagoj Bradač and Nemanja Draganić for introducing me to the world of Ramsey theory, showing me numerous beautiful papers and suggesting several open conjectures that I (unsuccessfully) tried to work on and enjoyed thinking about immensely.

Introduction

Erdős asks us to imagine an alien force, vastly more powerful than us, landing on Earth and demanding the value of $r(5, 5)$ or they will destroy our planet. In that case, he claims, we should marshal all our computers and all our mathematicians and attempt to find the value. But suppose, instead, that they ask for $r(6, 6)$. In that case, he believes, we should attempt to destroy the aliens.

Joel Spencer

In 1928, the English mathematician Frank Plumpton Ramsey[34] submitted a paper named *On a Problem of Formal Logic*, solving a special case of the decision problem for first-order logic. In this paper, he proved a lemma that would prove to be a major result in combinatorics and is now a theorem that bears his name:

Theorem 1: Ramsey's Theorem

For any positive integers k and l , there exists an integer $r = r(k, l)$ such that if the edges of the complete graph K_r are coloured in either red or blue, there is either a red copy of K_k or a blue copy of K_l somewhere in the graph.

The numbers $r(l, k)$ are today called *Ramsey numbers*, and their study constitutes a large domain of combinatorics called Ramsey Theory. Erdős and Szekeres[18, 15] would later prove the first bounds on these numbers; but in the nearly hundred years since Ramsey's initial publication, the bounds have remained extremely far apart, despite several papers improving both bounds being published since. Notably, Erdős' first paper gave birth to the probabilistic method, which would expand into a large repertoire of useful tools that has been used to great effect in solving problems in Ramsey theory. Research into Ramsey theory also expanded to ask what happened when an arbitrary finite number of colours were permitted, as well as what occurred with infinite graphs, which later became a cornerstone of combinatorial number theory and additive combinatorics.

Today, there are two classes of Ramsey numbers that are well studied. The first is a simple generalisation of Ramsey's first lemma, which is to ask for the Ramsey number $r(H_1, H_2)$ of any two graphs H_1, H_2 rather than just complete graphs in particular. Of particular interest is the specific case where $H_1 = H = H_2$, simply called the Ramsey number $r(H)$, or $r(k) = r(k, k)$ for the original, complete graph case. A closely related question is that of finding the *size-Ramsey number* $\hat{r}(H)$, which asks for the minimum number of edges the host graph must have instead of vertices, if we no longer force it to be a complete graph.

The second class of Ramsey numbers are the *induced-Ramsey numbers*, which ask for the minimum number of vertices or edges $r_{ind}(H), \hat{r}_{ind}(H)$ such that there is an induced monochromatic copy of the graph H for any colouring. These numbers are significantly harder to bound and their existence alone was a difficult problem that was not solved until the 1970s.

1 Ramsey numbers

The complete graph with 6 points and 15 edges has each edge colored red or blue. Show that we can find 3 points such that the 3 edges joining them are the same color.

William Lowell Putnam Mathematical Competition, 1953

Ramsey's original lemma proved only the existence of his eponymous numbers. We begin by discussing the bounds and proofs for these that have appeared over the years. For the entirety of this paper, unless specified otherwise, bounds given only hold true for sufficiently large n and not in all generality. We will also assume the graphs in question are all connected. We also adopt the convention that $r(G, H)$ refers to the smallest number r where a colouring of K_r either contains a **red** copy of G or a **blue** copy of H ¹. Unless mentioned explicitly otherwise, a colouring of a graph is in two colours.

We also use the following standard definitions. Given a graph G and subsets $S, T \subseteq V(G)$ of the vertices, then

- $|G|$ or the *size* of G denotes the number of edges of G , whilst the *order* of G is the number of vertices of G ;
- $G(S) \subseteq E(G)$ denotes the set of edges having at least one endpoint in S ;
- $G[S]$, or the *induced subgraph* denotes the set of edges having both endpoints in S and by abuse of notation this can refer to either the edge set or the graph;
- $\Gamma_G(S) \subseteq V(G)$ denotes the set of vertices adjacent to at least one element of S ;
- $N_G[S] := \Gamma_G(S) \cup S$ or the *closed neighbourhood* of S denotes the set of vertices adjacent to at least one element of S or in S .
- $N_G(S) := \Gamma_G(S) \setminus S$ or the *open neighbourhood* of S denotes the set of vertices adjacent to at least one element of S whilst not being in S .

Unless specified otherwise, *neighbourhood* will refer to the open neighbourhood.

We say, for two graphs G, H , that G is *H-Ramsey* if, for every colouring of G with two colours, one obtains a monochromatic copy of H .

1.1 Diagonal Ramsey numbers

Erdős and Szekeres[18] proved the first explicit bounds on the Ramsey numbers.

¹This is not the standard in literature - usually the first graph is blue and the second is red.

Bound 1: Erdős-Szekeres upper bound 1935.

$$r(k, k) \leq \binom{2k-2}{k-1}.$$

Proof. Firstly, note that clearly $r(2, k) = k$ as either we have a red edge or the entire graph is blue. Also, $r(x, y) = r(y, x)$ is evident.

Lemma 1.1. $r(k, \ell) \leq r(k-1, \ell) + r(k, \ell-1)$.

Take the complete graph $G = K_{r(k-1, \ell) + r(k, \ell-1)}$ and colour the edges red and blue in any manner. Consider any vertex $v \in V(G)$ and partition the remaining vertices into two sets R, B where

$$R = \{ u \in V(G) \setminus v \mid (u, v) \text{ is red} \}$$

with a similar definition for B and the colour blue. Either $|R| \geq r(k-1, \ell)$ or $|B| \geq r(k, \ell-1)$. In the first case, either the subgraph induced by R has a blue copy of K_ℓ , or a red copy of K_{k-1} all connected to v with red edges, yielding a red copy of K_k in the original graph. The other case proceeds in the same manner.

This lemma along with our initial observation shows the bound

$$r(k, \ell) \leq \binom{k+\ell-2}{k-1}$$

by induction. In particular, $r(k, k) \leq \binom{2k-2}{k-1} \leq 4^k$. □

Finding a lower bound, however, was not as simple; providing an explicit construction was often tricky and the most elegant ones were often far from optimal. For example, the bound $R(k, k) \geq (k-1)^2$ was easy to prove by splitting the vertices into $k-1$ classes of equal size and then colouring them based on whether the edge was within a class or between two classes, but seemed to be far from the true lower bound. In 1947, Erdős[15] found a novel proof for a much better bound that was one of the first examples of the probabilistic method.

Bound 2: Erdős lower bound 1947.

$$r(k, k) \geq 2^{k/2}.$$

Proof. Let A denote the event that we colour the complete graph $K_{2^{k/2}}$ with two colours in such a manner that there are no monochromatic copies of K_k . The key idea will be that if we write A^c as a union of “bad” events, whose total probability is less than 1, then the probability of A is at least 0 and some colouring exists with this property.

Let $N \leq 2^{k/2}$. Consider the complete graph $G = K_N$ on N vertices and colour it randomly, with each edge being either red or blue with equal probability and independently

of the other edges. Consider any subset $U \subset V(G)$ of k vertices; the probability that they all have the same colour is clearly $2^{1-k(k-1)/2}$ (two choices of colour, and then a probability of $1/2$ for every edge to be that colour, with the edges being independent). This is a bad event as described above. There are $\binom{n}{k}$ possible bad events, so we can bound the total probability of bad events by $\binom{n}{k} 2^{1-k(k-1)/2}$. Note in particular that this bound is “naive” in the sense that it does not account for the fact that many of the events are dependent on each other.

Now note that

$$\begin{aligned} \binom{n}{k} 2^{1-k(k-1)/2} &\leq \binom{2^{k/2}}{k} 2^{1-k(k-1)/2} \\ &\leq 2^{k^2/2} / k! \cdot 2^{1-k(k-1)/2} \\ &\leq 2^{k/2+1} / k! \\ &\leq 1 \end{aligned}$$

where the last inequality is easy to prove for $k \geq 3$. □

This idea of providing nonconstructive lower bounds by proceeding randomly is the backbone of the probabilistic method. The final inequality can be improved using Sterling’s formula for binomial coefficients to $r(k) \geq k2^{k/2}(\sqrt{2}/2e + o(1))$.

Nevertheless, the lower bound would stand untouched for 27 years, until Spencer[38] attacked the problem again using a novel new tool from the probabilistic method repertoire: the Lovász Local Lemma.

Bound 3: Spencer lower bound 1975.

$$r(k, k) \geq k2^{k/2}(\sqrt{2}/e + o(1)).$$

Proof ideas: We will not do the proof entirely, but rather explain the motivation and why the local lemma improves the bound. Firstly, the lemma:

Lemma 1.2. *Let $A_1, A_2, \dots, A_k$ be a sequence of events such that each event occurs with probability at most p and such that each event is independent of all the other events except for at most d of them. If*

$$4pd < 1$$

then there is a nonzero probability that none of the events occurs.

Essentially, the lemma says that if there is a bound on the dependence between events, then we can still conclude that none of them occur. This is much more powerful than the union bound employed by Erdős 27 years prior.

Applying the bound is particularly desirable because “bad” events in this scenario are dependent on relatively few other events. In particular, given two sets of vertices $U, W \in V(G)$, if $|U \cap W| < 2$, they have no common edges. For a set $S \in V(G)$ of size k , let

the event that all the edges $G[S]$ have the same colour be denoted by A_S . We get that the number d of such events not independent with a given A_S is simply the number of choices of a set T whose intersection with S is greater than 2. This can be bounded by

$$d \leq \binom{k}{2} \binom{N}{k-2}$$

where N is the number of vertices of the graph, with $p = 2^{1-k(k-1)/2}$ as before. We therefore want to take the maximal N such that $4pd < 1$; applying Sterling's formula would then give us the desired bound. $\square$

Spencer remarked that by examining the asymptotics in the local lemma, one could further conclude that almost all colourings were good for values slightly smaller than Erdős' lower bound, but that this was not the case for his own bound. He also applied the lemma to the general case to try and derive a bound on $r(k, \ell)$, which we will mention in the next section. Spencer would go on to improve the local lemma itself, from necessitating $4pd < 1$ to $ep(d+1) < 1$, and use this to improve his bound further.

An interesting question to ask oneself is what is the best deterministic lower bound we can get, by providing explicit constructions. Frankl and Wilson[24] found graphs with 2^n vertices that were $2^{\Omega(n)}$ -Ramsey (meaning that they were between 2^{cn} -Ramsey and 2^{Cn} -Ramsey for two constants $c < C$). Note that this is only a polynomial bound, and much weaker than the probabilistic bounds we have been discussing. In recent years the constructive lower bound has been improved significantly using methods from computer science, but these are far beyond the scope of this paper.

The upper bound, meanwhile, was pushed down several times. Firstly, Graver and Yackel[27] improved it slightly by considering the degrees of vertices in Ramsey graphs and using an extremal argument to give a lower bound on the number of edges. The first significant improvement, however, came from Graham and Rödl[26], who managed to divide Erdős' bound by a factor of $\log \log k$.

Bound 4: Graham-Rödl upper bound 1987.

$$r(k, k) \leq 6 \binom{2k-2}{k-1} / \log \log k.$$

Ideas from the proof will be discussed in the next section, as they showed a more general result on the Ramsey number $R(k, l)$.

After this significant leap forward, Thomason[39], Conlon[8] and Sah[37] would all go on to make progress on the upper bound.

Bound 5: Thomason upper bound 1988.

$$r(k, k) \leq \binom{2k-2}{k-1} / k^{1/2-C/\sqrt{\log k}}.$$

Thomason’s method was essentially a refinement of Graham and Rödl’s method, but with tighter functions in the inductive step, allowing him to get a tighter bound. In particular, he applied Goodman’s formula, a result on the number of monochromatic forced triangles.

Bound 6: Conlon upper bound 2009.

$$r(k, k) \leq \binom{2k-2}{k-1} / k^{C \log k / \log \log k}.$$

Conlon expanded upon Thomason’s work by introducing the notion of quasirandomness, which had been first discussed by Thomason under the term ”jumbledness”. In essence, a quasirandom graph is a graph that shares certain properties with random graphs with high probability. Conlon used the fact that counterexamples to Ramsey (graphs containing neither red K_r nor blue K_l) were quasirandom and therefore was able to circumvent Goodman and use stronger results for more general cliques and not just triangles.

Bound 7: Sah upper bound 2020.

$$r(k, l) \leq \binom{2k-2}{k-1} / k^{C \log k}$$

Sah’s proof was an optimisation of quasirandomness to its natural limit; Conlon himself stated that he felt the method had nothing more to give.

However, the factor of improvement throughout this period was negligible compared to the bound itself, which was exponential. It was not until this year that the first exponential progress in nearly a century was made by Campos, Griffiths, Morris and Sahasrabudhe[6].

Bound 8: Campos et al. upper bound 2023.

$$r(k, k) \leq 3.993^k$$

This result was a an unprecedented exponential improvement on the upper bound. The proof is quite complex and beyond the scope of this paper, but we note that it used completely new methods from those that came before it.

1.2 Off-diagonal Ramsey numbers

Ramsey’s original theorem discussed the assymmetric case where he sought two different complete graphs in red and blue. Spencer[38] gave two bounds in his 1975 paper: the first, essentially a variant of Erdős’ probabilistic method, the second using a generalisation of the local lemma.

Bound 9: Spencer lower bound 1975.

Asymptotically, we have

$$r(k, (2 \ln n) n^{2/(k-1)} (1 + o(1))) > n$$

and in particular, for k fixed, $l \rightarrow \infty$, we have

$$r(k, \ell) > \ell^{(k-1)/2+o(1)}.$$

Proof ideas: Again, we just mention the ideas. This time, rather than taking red and blue randomly with probability $1/2$, we make the probability a variable p that we will fix later, making edges red with probability p and blue with probability $1 - p$. As k and ℓ may now be different, we are willing to have more edges in the colour with the larger of the two values. Notably, if we know that there exists n, p such that

$$\binom{n}{k} p^{\binom{k}{2}} + \binom{n}{\ell} p^{\binom{\ell}{2}} > 1$$

then $r(k, \ell) > n$, by the union bound. Optimising for best value of p that maximises n , we get the above results. $\square$

What happens now if we try and use the local lemma rather than the union bound?

Bound 10: Spencer lower bound 1974.

Asymptotically, for k fixed, $l \rightarrow \infty$, we have

$$r(k, \ell) > \ell^{((\binom{k-2}{2}-2)/(k-2)+o(1))}.$$

This follows from the same idea as before but this time using a generalisation of the local lemma rather the union bound.

As mentioned before, the next progress came from Graham and Rödl[26], who proved the following bound in their paper.

Bound 11: Graham-Rödl upper bound 1987.

$$r(k, \ell) \leq 6 \binom{k+\ell}{k} / \log \log(k+\ell).$$

Proof ideas: Let us discuss the a few ideas of the proof.

The first idea is to bound the minimum number of monochromatic triangles as a function of the proportion of the number of edges that are red. Consider $S \subseteq V(G)$, $|S| = 3$. If $G[S]$ is not monochromatic then it must contain two red and one blue edge, or vice versa. In particular, there are exactly two vertices where we have a red edge meet a blue one. Notably, this motivates a double counting argument: one can see that that we can also count the number of instances where a red edge meets a blue one at a given vertex v as

the product $r_v b_v$ where r_v and b_v denote the red and blue degrees of v , respectively. It then follows that

$$\# \text{ of monochromatic triangles} = \binom{n}{3} - \sum_{v \in V(G)} r_v b_v$$

Of course, if nearly all or almost none of the edges are red, then we have a huge amount of monochromatic triangles; the minimal case occurs when we have an equal number of red and blue edges, and the inequality we obtain will reflect this, as we can set β to be the proportion of edges that are red, and then use this to find an upper bound on the number of removed triangles described above. Glossing over the calculation itself, we obtain that the number of triangles is at least

$$(\beta^3 + (1 - \beta^3)) \binom{n}{3} - \beta(1 - \beta) \binom{n}{2}.$$

Our motivation now is to use the fact that if we have enough monochromatic triangles, we will know that we have cliques. In particular, if a red edge is in $r(k-2, l)$ triangles, we can finish, and similarly for blue edges. The reason is that, considering the endpoints of these triangles, either we find a red copy of K_{k-2} connected to a red copy of K_2 with only red edges (which gives a red copy of K_k) or a blue copy of K_l .

The next step is to establish a similar bound for the case when one of the two values k, l is significantly larger than the other. If we assume WLOG $k > el^2$, we can show

$$r(k, l) < 5 \binom{k+l}{l} / \log(k/el^2).$$

This follows from Erdős and Szekeres's lemma being applied repeatedly to bound the LHS as a sum of smaller Ramsey numbers. In particular, this sum can now be attacked with the bound

$$r(k, 3) < k^2 / \log(k/e)$$

which follows from the fact that in triangle-free graphs, the neighbourhood of any vertex is an independent set; with a large enough size, one can guarantee an independent set of size k . This specific bound, however, is not trivial to see and depends on the lemma that for a graph G with n vertices and with average degree d ,

$$\alpha(G) > n(d \log d - d + 1) / (d - 1)^2.$$

Admitting this lemma as true (the proof, which is by induction on n , but we will not discuss it here) we have that, if we have some graph with no red copy of K_k or blue copy of K_3 , then we know the degree of every vertex in the blue graph is at most $k-1$. This implies the average degree is at most $k-1$ and so using the lemma, we get

$$(k-1) > n(k \log k - k + 1) / (k-1)^2.$$

and after rearranging we get the desired bound on $R(k, 3)$, which in turn can be used to show the bound on $r(k, l)$ for the case where they are quite different.

We omit the most involved part of the proof, which is to now apply a recursive argument where, for sufficiently large k, l , $r(k, l)$ is bounded in function of either $r(k - 1, l)$, $r(k, l - 1)$, $r(k - 2, l)$ or $r(k, l - 2)$. Combining these with the above results, we can express $r(k, l)$ in terms of several applications of the recursive argument along with one of our many “base cases”. $\square$

The bound on $r(3, k)$ used by Graham and Rödl has since been significantly improved. The current best bounds were found by Ajtai, Komlós and Szemerédi[1] and Kim[30]. In particular, the asymptotic order is known.

Bound 12: Ajtai and Komlós and Szemerédi lower bound 1980 and Kim upper bound 1995.

$$C(1 - o(1))k^2 / \log k \leq R(k, 3) \leq (1 + o(1))k^2 / \log k.$$

In their respective papers, Thomason[39], Conlon[8], Sah[37] and Campos, Griffiths, Morris and Sahasrabudhe[6] all proved better bounds on the off-diagonal Ramsey numbers as well. We list these below as well.

Bound 13: Thomason upper bound 1988.

If $l \leq k$,

$$r(l, k) \leq \binom{k + l - 2}{k - 1} / e^{l \log k / 2k - C \sqrt{\log k}}.$$

Bound 14: Conlon upper bound 2009.

Let s and t be fixed positive constants with $t \leq 1$. Then there exists a constant $C_{s,t}$ such that for $k \geq l \geq tk$,

$$r(k, l) \leq C_{s,t} \binom{k + l - 2}{k - 1} / e^{sl \log k / k}.$$

Bound 15: Sah upper bound 2020.

For each $t \in (0, 1/2)$ there is a constant $c_t > 0$ such that

$$r(k, l) \leq \binom{k + l - 2}{k - 1} / e^{c_t (\log k)^2}$$

for $t \leq l/k \leq 1$ and $l \geq 1/c_t$.

Bound 16: Campos et al. upper bound 2023.

For $k \geq l$ there exists $t > 0$ such that

$$r(k, l) \leq \binom{k+l-2}{k-1} / e^{\delta l + o(k)}.$$

1.3 Multicoloured Ramsey numbers and paths, cycles and trees

Multicoloured Ramsey numbers are much-less studied than their bicolored counterparts. Indeed, until 2016, the only known nontrivial multicolour Ramsey number was $R(3, 3, 3) = 17$. One can remark the simple result

$$R(k_1, \dots, k_r) \leq 2 - r + \sum_{i=1}^r R(k_1, \dots, k_{i-1}, k_i - 1, k_{i+1}, \dots, k_r)$$

that follows from the same pigeonhole argument first used by Erdős and Szekeres.

As mentioned in the introduction, it is quite simple to provide an analogous definition of Ramsey numbers for arbitrary graphs. A trivial upper bound for the Ramsey number $r(G, H)$ of two graphs is just $r(|V(G)|, |V(H)|)$, and a lower bound is $r(\omega(G), \omega(H))$. More simply put, the partial order induced by the subgraph relationship allows us to find simple (but usually weak) upper and lower bounds from existing ones.

It is tempting to conjecture that there is a positive correlation between the number of edges or vertices in a graph and its Ramsey number. Whilst this is clearly true globally, it is clearly not a strict relationship: consider for example the star $K_{1,n}$. It is easy to see that $r(K_{1,n}, K_{1,n}) = 2n$ or $2n+1$; a monochromatic copy of the star exists if and only if there is a vertex with degree n in either colour. The graph K_{2n+1} can be coloured in such a fashion that every vertex has degree $n+1$ in both colours if and only if n is even (if n is even then we decompose the graph into n Hamiltonian cycles and colour half in each colour; if n is odd then the handshaking lemma easily shows that not every vertex can have an odd degree in both colours). If n is even, then it is easy to see each vertex in K_{2n} is degree n in some colour. This implies that the Ramsey number of the star is linear in its number of vertices and edges; as seen in the previous section, for the complete graph it is superpolynomial in both.

There are a huge wealth of results in this domain. However, there are far too many interesting results to discuss them all, so we will focus on results about the Ramsey number of familiar graphs, and in particular those results where the value is known exactly and the proof is combinatorially appealing. The first concerns trees and complete graphs and was published by Chvátal[7].

Result 1: Tree versus Complete Graph due to Chvátal.

For any tree T_n on n vertices,

$$r(T_n, K_m) = 1 + (n - 1)(m - 1)$$

Proof. For the lower bound, just colour $(m - 1)$ disjoint copies of K_{n-1} in red and the remaining edges (which just form a Turán graph) in blue. Clearly, the red edges do not form any connected graph of size larger than n , and any subset of at least m vertices contains at least two from the same equivalence class (as endowed by the red edges) which therefore do not share a blue edge. Now consider any colouring of $K_{1+(n-1)(m-1)}$. Assuming there are no blue copies of K_m , the red graph is at least n -chromatic (there are no independent sets larger than size $m - 1$ and each colour defines an independent set).

We now prove the following lemma, due to Dirac[11]:

Lemma 1.3. *A finite k -chromatic graph admits a graph of minimum degree $k - 1$ as a subgraph.*

Proof. Let G be a finite k -chromatic graph. Remove vertices one by one. If the chromatic value decreases to $k - 2$; rewind a step and remove another vertex. Terminate when this is not possible (i.e, removing any vertex would decrease the chromatic number). This must happen at some point, as the G initially has finite size and if the size is eventually smaller than k then it is at most $k - 1$ chromatic. The preceding graph is said to be *critical*; it is $k - 1$ chromatic but removing any vertex would make it $k - 2$ chromatic.

The graph thus obtained must have minimal degree at least $k - 1$. If we assume that there is a vertex v with degree less than $k - 1$, we can remove this vertex and colour the remaining vertices with $k - 1$ colours; v can then be coloured in one of these as well since it is only restricted from at most $k - 2$ colours, a contradiction. $\square$

Of course, the same result follows for graphs that are more than k -chromatic and so our red graph contains a subgraph with minimum degree $k - 1$. But this subgraph contains *every* tree on k vertices. This can be seen by induction: the case $k = 2$ is trivial. Now, for a tree T of arbitrary size n , remove a leaf v previously attached to some u to give a tree T' : by induction, T' is contained in every graph of minimum degree $k - 2$, including our graph with minimum degree $k - 1$. If we now find a copy of T' , then the vertex corresponding to u is connected to at least one vertex not already in the tree, which we can take as v . Note that stars are trees, meaning $r(K_{1,n}, K_m) = 1 + n(m - 1)$. $\square$

The next one is for cycles, and was found by Faudree and Schelp[20].

Result 2: Cycle versus Cycle due to Faudree and Schelp.

- If $3 \leq m \leq n$, $(m, n) \neq (3, 3)$, m odd, then $r(C_m, C_n) = 2n - 1$.
- If $4 \leq m \leq n$, $(m, n) \neq (4, 4)$ and both are even then $r(C_m, C_n) = n + m/2 + 1$.

- If $4 \leq m < n$, m odd and n even, $r(C_m, C_n) = \max(n + m/2 + 1, 2s + 1)$.

Proof ideas: We discuss a couple of elegant ideas from the proof. It begins by stating several results on sufficient conditions for cycles to occur. These include a result on the minimum number of edges as a function of some l and $n = |V(G)|$ to have a cycle of length l ; on the minimum number of edges as a function of n to have cycles of all lengths from 3 to $(n + 3)/2$; on $r(C_n, C_n)$ being bounded above by $2n - 1$; on known values when one of the cycles is C_3, C_4 or C_5 (which was already completely solved) and on the fact that if any two non-adjacent vertices always have total degree sum at least n then either the graph is complete bipartite or all cycle lengths are present in the graph.

One key idea is that if a vertex v is connected to more than half the vertices in a cycle $C_L = x_1, x_2, \dots, x_L$ of length L , then together they contain cycles of every possible length. This is true by pigeonhole over the pairs of vertices in the cycle of distance i : some pair x_j, x_{j+i} are both connected to v and therefore together form a cycle of length $i + 2$, and i can freely be taken between 1 and $L - 1$.

Another key idea is that if the red edges contain a cycle $C_L = x_1, x_2, \dots, x_L$ of length L and the blue edges contain no cycle $C_l = y_1, y_2, \dots, y_N$ of length $N \leq L$, then for every M between the two, there is either a red cycle of length M or $M + 1$. The proof is by downwards induction starting at a cycle C of length L and proving there is a cycle of length $L - 1$ or $L - 2$. If this is not the case, we know that there are no red edges crossing a distance of length 2 or 3 in C . By using these blue edges to create a cycle, we can find a cycle of length N , contradiction.

Once these two ideas have been established then the rest of the proof, along with the previous results, is more or less a sequence of deductions and bashing out different cases. $\square$

Faudree and Schelp would also collaborate with Lawrence and Parsons[22] to attack the problem of cycles versus paths.

Result 3: Cycle versus path due to Faudree and Lawrence and Parsons and Schelp

- If $3 \leq m \leq n$, m odd, then $r(P_n, C_m) = 2n - 1$.
- If $4 \leq m \leq n$, m even, then $r(P_n, C_m) = n + m/2 - 1$.
- If $2 \leq m \leq n$, n odd, then $r(C_n, P_m) = \max(n + \lfloor m/2 \rfloor - 1, 2m - 1)$.
- If $2 \leq m \leq n$, n even, then $r(C_n, P_m) = n + \lfloor m/2 \rfloor - 1$.

We will not discuss the proof as it is very similar in flavour to the previous one and mostly a case of working out the different cases. Notice that the previous results are trivially upper bounds here and coincide with the values sometimes (meaning you are sure to have a path if and only if you are sure to have a cycle).

The first to settle the exact Ramsey number for two paths were Gerencsér and Gyárfás[31].

Result 4: Path versus path due to Gerencsér and Gyárfás

For $k \geq l$ we have

$$r(P_k, P_l) = k + \lfloor (l+1)/2 \rfloor.$$

Proof. We proceed via induction on k ; the base case $k = 2$ is simple. Consider a two-colouring of a complete graph G on $r := k + \lfloor (l+1)/2 \rfloor$ vertices. Suppose the longest red path is of length $k - 1$ and let $U = \{u_1, u_2, \dots, u_k\}$ denote the vertices of such a path (in order), and label the remaining vertices $V = \{v_1, v_2, \dots, v_m\}$ where $m = \lfloor (l+1)/2 \rfloor$ in any order.

Note that for any two consecutive vertices in U , if they are both connected to a vertex in V by a red edge we could extend the path, so they together are connected to each vertex in V by red edges at most once. Furthermore, the edges from u_1 and u_k to V must all be blue for the same reason.

Consider now a maximal blue path with the property that it contains neither u_1 nor u_k and that it starts in V and alternates between V and U , also finishing in V . If it contained all points in V then we could extend it by 2 (using u_1, u_k) to create a path of length l . Suppose instead this is not the case and let $W \subseteq V$ be the points it does not contain. Consider a second maximal path, similar to the one just defined but using W instead of V .

Every point of V must be in one of the two paths. Suppose this was not the case; there is a vertex v not in either path and as each path has one more vertex in V than in U and $|V|$ is smaller or equal to half of $|U|$, by pigeonhole there are two consecutive points u_i, u_{i+1} not in either path. However, taking two of the endpoints of the two paths along with v , by pigeonhole either u_i or u_{i+1} is connected to two of them, allowing us to extend the paths, contradiction.

Now connect these two paths using u_1 and u_k to form a circuit. For odd l this contains a path of length at least l ; for even l the circuit is length l but we observe that the circuit does not more than half the elements of U and again, taking two consecutive vertices not present, one of them must be attached to the circuit, giving a path of length l .

The lower bound is left as an exercise for the reader, but can be deduced from the proof. $\square$

Finally, we mention a bound on the multicoloured number of the tree, found by Erdős and Graham[16]

Result 5: Multicoloured tree due to Erdős and Graham

$$(n-1)k + 1 < r^k(T_n) < 2kn + 1$$

We will not discuss the lower bound. The upper bound simply follows from the fact that any graph with average degree $2n$ contains a copy of T_n , which is a nice brainteaser suggested to the reader if they do not know the proof.

1.4 Conjectures

It is strongly believed that $r(5, 5)$ is 43; for now, we know the value is between 43 and 48. Other conjectures include that $r(K_{2,n}, K_{2,n})$ is either $4n - 3$ or $4n - 2$, that $K(C_n, K_m) = (n - 1)(m - 1) + 1$ except certain small values; that $r(T_n)$ is bounded above by $2n - 2^2$ and the stronger case that $r(T_n, T_m) \leq n + m - 2$.

²Note that this is less than or equal to the bound for the star, implying the star is the worst case of all trees. The even case as well as the case for large numbers have both been proven.

2 Size-Ramsey numbers

Another roof, another proof.

Paul Erdős

In 1978, Erdős, Faudree, Rousseau and Schelp[19] were the first to ask the question of the minimum number of edges rather than vertices needed to guarantee a monochromatic copy of a given graph. For some graphs, such as the star $K_{1,n}$, the number of edges is very low, as remarked in the previous section, since the star $K_{1,2n-1}$ suffices. For others, such as the complete graph, it has been proven that the best choice of host graph is itself a complete graph. Studying the size-Ramsey number adds an added layer of difficulty because determining the best host graph is often nontrivial.

2.1 Complete graphs

The first result, published by Erdős, Faudree, Rousseau and Schelp in their aforementioned paper, was actually communicated to them by Chvátal.

Result 6: Size-Ramsey number of complete graph due to Chvátal.

$$\hat{r}(k, l) = \binom{r(k, l)}{2}$$

and the only graph of this size that induces K_k or K_l is the complete graph on $r(k, l)$ vertices.

Proof. Let $r := r(k, l)$.

Firstly, consider a an arbitrary non-complete graph G and a pair of vertices u, v that are not adjacent, and colour the edges arbitrarily. If neither $G \setminus u$ nor $G \setminus v$ contain a red copy of K_k or a blue copy of K_l , then G does not either. The reason is obvious: a monochromatic copy cannot use both u and v because of the missing edge.

Now we will show that any graph of size at most $\binom{r}{2}$ that must have a red K_k or blue K_l must contain K_r as a subgraph, implying what we want to show. Take an arbitrary graph G such that $\omega(G) < r - 1$. Clearly, the graph can be decomposed into (possibly nondisjoint) maximal cliques of size at most $r - 1$ and our motivation will be to colour them individually in such a manner that they do not contain a red K_k or blue K_l (which can be done by definition) and so they coincide over their intersections, meaning our previous observation would allow us to finish. A natural approach would be to try and "reflect" over non-edges (so for example, if we had $G = K_r$ with one edge missing between u, v then we could colour $G \setminus u$ with a good colouring and then colour the edges from v in the same colour as those from u .)

We will apply this idea via induction on the order of the graph. Suppose it holds true that for all non-complete graphs of order less than p and size at most $\binom{r}{2}$. Clearly, it holds for $p < r$, completing the base case, so now suppose $p \geq r$ and take any graph G satisfying the conditions. Consider any two non-adjacent vertices u, v . We can make the situation symmetric by adding edges between u and $N_G(v)$ and vice versa, giving a new graph G' . This means that if we can now colour $G' \setminus u$ we are done, as we can equip the same colouring to $G' \setminus v$ with it coinciding over $G \setminus \{u, v\}$. Additionally, we know that as since G has at most $\binom{r}{2}$ edges, so does $G' \setminus u$ (as we lose and gain exactly one edge for every edge adjacent to u). Moreover, the order of this graph is $p - 1$, meaning the only thing remaining from our induction hypothesis is that G' may be complete. If it is not complete, we can finish by induction.

Therefore, let us proceed in the case where G' is complete. This means $G' \setminus \{u, v\} = G \setminus \{u, v\}$ must be a copy of R_{r-1} and every vertex $w \in V(G)$ is connected to exactly one of u, v (at least one so that they both end up connected to it, and at most one to satisfy the initial bound on the total number of edges). We also know they are both connected to at least one vertex initially. In other words, $V(G \setminus \{u, v\})$ is the disjoint union of $N_G(v)$ and $N_G(u)$.

This implies there is a vertex w in $N_G(v)$ that is not connected to u but to one of its neighbours; if we take this vertex instead of v initially, then we can conclude with the above argument because the neighbourhoods will not give a disjoint partition this time, preventing the issue. $\square$

One thing to note is that the proof relies on the graph having at most $\binom{r(k,l)}{2}$ edges; it is interesting to ask whether the stronger result that “ G must contain $R_{r(k,l)}$ as a subgraph” holds. The answer is in the negative (and was shown to hold for the much stronger forbidden subgraph $R_{\max(k,l)+1}$), as proven by Nešetřil and Rödl[33] in 1976. The proof is beyond the scope of this paper. Consider instead the following construction found during a conversation between David Rusch and the author for a K_6 -free graph inducing K_3 .

Result 7: Weaker case of Nešetřil and Rödl’s result for a small k .

There is a graph G that is K_3 -Ramsey but does not contain K_6 as a subgraph.

Proof. Remark the fact that there are finitely many distinct colourings of K_5 that do not contain a monochromatic triangle; indeed, each such colouring corresponds to a cyclic ordering of the vertices and so there are exactly 24 such colourings, each of which is composed of a red and blue cycle of length 5.

Now consider a colouring of the Turán graph $T(k, 5)$, $k \geq 5$ without a monochromatic triangle. We claim all edges between the elements of two different independent sets have the same colour. To see this, consider A, B, C, D, E each taken from one of the 5 different maximal independent sets. The complete subgraph induced by these 5 must be one of the 24 colourings described before. Removing A , we remove two red edges and two blue edges and note that the graph on the remaining four vertices must have a blue and a red

path of length 3. Adding any other element A' from the same maximal independent set as A , we see that the edges between A' and the other four vertices are forced in order to again create the red and blue cycle of length 5 that characterises a triangle-free colouring of K_5 .

Now we admit that there are constructions of graphs with arbitrarily large girth and chromatic number; this is an old result known due to Erdős. Take a graph G with chromatic number 25 but girth 4. We use the structure of G to construct the desired graph.

To each of the vertices of G , assign a copy of the Turán graph $T(9, 5)$. Glue the singleton from each graph together (in other words, they all share exactly one vertex - call it Z) to obtain a pseudo-book graph and label the other eight $A_v, A'_v, B_v, B'_v, C_v, C'_v, D_v, D'_v$ where (X_v, X'_v) is one of the four missing edges for the $T(9, 5)$ corresponding to $v \in V(G)$. Now, for every edge $u, v \in G$, introduce the 16 edges (X_u, Y_v) for $\{X, Y\} \neq \{A, C\}$ or $\{B, D\}$, and the 16 edges (X'_u, Y'_v) for $\{X, Y\} \neq \{A, B\}$ or $\{C, D\}$. We claim this graph cannot be coloured without creating a monochromatic triangle.

Take an arbitrary colouring of the graph. Firstly, if any of the $T(9, 5)$ has a monochromatic triangle, we are done, so we may assume each “page” has one of the 24 possible triangle-free colourings. This induces a colouring in 24 colours of the vertices of G via the page-vertex correspondence, so as $\chi(G) = 25$ there are two adjacent vertices u, v with the same colour.

Consider the subgraph induced by the two corresponding pages. We already know $(Z, X_u), (Z, X'_u), (Z, X_v), (Z, X'_v)$ have the same colour for $X \in \{A, B, C, D\}$. Moreover, we know that two of $\{(Z, A_u), (Z, B_u), (Z, C_u), (Z, D_u)\}$ are red and two are blue. We now have three cases to consider.

- If (Z, A_u) and (Z, B_u) are the same colour, then $\{(Z, A_u), (Z, A_v), (Z, B_u), (Z, B_v)\}$ are the same colour and all edges between A_u, A_v, B_u, B_v are present, making a monochromatic triangle obligatory.
- If (Z, A_u) and (Z, C_u) are the same colour, then $\{(Z, A'_u), (Z, A'_v), (Z, C'_u), (Z, C'_v)\}$ are the same colour and all edges between A'_u, A'_v, C'_u, C'_v are present, making a monochromatic triangle obligatory.
- If (Z, A_u) and (Z, D_u) are the same colour, then $\{(Z, D_u), (Z, D_v), (Z, D_u), (Z, D_v)\}$ are the same colour and all edges between D_u, D_v, D_u, D_v are present, making a monochromatic triangle obligatory.

Finally, we note that this graph does not contain any copy of K_6 . It suffices to show that the graph without Z has no copy of K_5 , and we remark it is just a collection of pages (corresponding to vertices) that are all copies of $T(8, 4)$, with some additional edges between them based on the above rule. Consider any 5 other vertices.

- If they are all from the same page, then trivially they cannot induce K_5 .

- If they are from two different pages, then first note that there are no edges between X_u and Y'_v for $u \neq v$, $X, Y \in \{A, B, C, D\}$, so they must all be of the form X or X' but not a mixture. But in both cases, the induced graph of those eight vertices is isomorphic to $T(8, 4)$ and does not contain K_5 .
- If they are from three different pages, then as G is K_3 -free, they cannot all be connected to each other.

We have therefore shown the existence of a K_3 -Ramsey graph not containing K_6 as a subgraph. Note that the stronger result of Nešetřil and Rödl would imply the existence of a K_3 -Ramsey graph not containing K_4 as a subgraph. $\square$

2.2 Paths and bounded-degree trees

Beck[2] was the first to prove a bound on the induced-Ramsey number of paths. His proof discusses the notion of *expander* graphs, which are graphs where all small sets of vertices have neighbourhoods (defined as those vertices adjacent to the set in question but not in the set itself) that are of a minimum size. Instead of discussing Beck's proof, we will instead provide a new proof heavily inspired by both Beck's original proof and also Draganić, Glock and Krivelevich's proof[12] for the induced size-Ramsey number of paths.

Bound 17: Linearity of Size-Ramsey number of paths due to Beck 1983.

$$\hat{r}(P_n) < 900n$$

Proof. We will instead prove the stronger bound $\hat{r}(P_n) < 250n$.

We would like to apply the DFS algorithm here. The algorithm, which is used to search a graph and has a variety of applications, works as follows:

- The algorithm is motivated by the concept of exploring a graph by going as far down a branch as possible before backtracking to the most recent point at which an unexplored branch was available, and proceeding down that branch.
- Our input will be a graph $G = (E, V)$ and a vertex $v \in V$. We will work with three sets - a set of “unexplored” vertices R , a set of vertices M which have been explored along with all their neighbours, and a stack P of vertices, which is a data structure where we add and remove from the top of the stack - in other words, the first vertex to leave is the most recent arrival. Initially, v is in P and every other vertex is in M .
- At every stage, we look at the top of the stack P and if the vertex there has at least one neighbour in R , we add one at random to the stack. Otherwise, we remove the vertex at the top of the stack P and add it to M .

Of particular interest to us is the fact that the stack, at any given moment, is just a path. It is also worth noting that every vertex in M has neighbours only in M or P . For this reason, we would expect that if we had a graph with expander properties, that at some point P is sufficiently large, since the neighbourhood of M is just P , and the size of M continually increases over time.

The first step is figuring out how to create a host graph with the property that, when coloured in two colours, one of the two always has expander properties. Taking a large random graph has a lot of benefits - many of the different aspects of the graph, like the degrees of the vertices, follow a distribution heavily centred around the mean.

How would we quantify the expander property in terms of events that are simple to assess probabilistically? We will take a leaf out of Beck's proof and adapt their choice of host graph. Notably, we want all sets of some size $m \leq n$ to have neighbourhood of size at least n . We will take $m = n$.

Lemma 2.1. *For any graph H there exists an induced subgraph G of H such that for every $S \subseteq V(G)$, the set of edges in G having at least one endpoint in S has cardinality at least $d(H) \cdot |S|/2$, where $d(H)$ denotes the average degree of H .*

Proof. Let X be a minimal (under set inclusion) induced subset of $V(H)$ for the property “The induced subgraph $H(X)$ has at least $d(H) \cdot |X|/2$ edges”. We propose that $G(X)$ is a viable choice for G . Suppose this is not the case - there is therefore a subset $S \subseteq V(X)$ such that the set of edges in X having at least one endpoint in S has cardinality less than $d(H) \cdot |S|/2$. However, this means the induced subgraph with vertex set $X \setminus S$ has at least $d(H) \cdot |X \setminus S|/2$ edges, contradicting the minimality of X . Note in particular $S = X$ is also impossible. $\square$

Let $G(S)$ denote, for a graph G , the set of edges having at least one endpoint in S . Let $G(X, Y]$ denote the edges having one endpoint in X and the other in $X \cup Y$. Let $N_G(S)$ denote the neighbourhood of S in G , i.e. all vertices not in S but adjacent to a vertex in S in the graph G .

Lemma 2.2. *Let F be a graph such that for every choice of disjoint sets $X, Y \subset F$ with $2 \leq |X| \leq n$ and $|Y| = |X|$, we have $|F(X, Y]| \leq d(F) \cdot |X|/4$. Then every subgraph $H \subseteq F$, $|H| \geq |F|/2$ contains a copy of P_n .*

Proof. Let H be an arbitrary subgraph of F containing at least half of the edges of F . By the previous lemma we know there exists a subgraph G of H such that for every $S \subseteq V(G)$ we have

$$|G(S)| \geq d(H) \cdot |S|/2 \geq d(F) \cdot |S|/4$$

We claim G contains a path of length n . Suppose firstly that G contains a subset X of size $m \leq n$ such that the neighbourhood $Y := N_G(X)$ of X has size $m' < m$. Then

$$|G(X)| = |G(X, Y)| \leq |F(X, Y)| < d(F) \cdot |X|/4,$$

contradiction with the assumption (Y does not have size m , but we can add arbitrary vertices from F to it until it does). Now, suppose instead that every subset S of G of size

$m \leq n$ has a neighbourhood of size at least m . In particular, G has order at least $2n$ - if it did not, then the neighbourhood of a set of size $\min(|V(G)|, n)$ would be too small (it clearly has at least 2 vertices, allowing us to use this argument).

Run the DFS algorithm as described above on G , and terminate when the set of explored vertices M is size n for the first time. As the stack P is the neighbourhood of M , it must contain at least n vertices, giving the desired path. $\square$

Now we want to show such a graph F exists. Let us take a random graph $G = \mathcal{G}(An, b/n)$ of order An where every edge occurs with independent probability b/n . We know by concentration inequalities that whp G has at least $(1 - o(1))A^2bn/2$ edges, so we may assume the average degree is at least Ac for any $c < b$.

In this case, a bad event is the existence of two sets X and Y with size $2 \leq m \leq n$ with more than $Ac m/4$ edges in $F(X, Y]$. The number of potential edges in $F(X, Y]$ is at most $3/2(m^2)$, and the number of edges we get is a binomial distribution. By a particular case of Chernoff³, we can say the probability there are more than $Ac m/4$ edges in this set is bounded by

$$\frac{6bme^{Ac m/4}}{nAc}.$$

We would like that

$$\binom{An}{m} \binom{An}{m} \left(\frac{6bme}{nAc} \right)^{Ac m/4} = o(1/n) \quad \forall 2 \leq m \leq n.$$

We can calculate

$$\begin{aligned} \binom{An}{m} \binom{An}{m} \left(\frac{6bme}{nAc} \right)^{Ac m/4} &\leq \left(\frac{Ane}{m} \right)^m \left(\frac{Ane}{m} \right)^m \left(\frac{6bme}{nAc} \right)^{Ac m/4} \\ &= \left[\left(\frac{Ane}{m} \right)^2 \left(\frac{6bme}{nAc} \right)^{Ac/4} \right]^m. \end{aligned}$$

We can assume b/c is arbitrarily close to 1.

$$\left[\left(\frac{Ane}{m} \right)^2 \left(\frac{6me}{An} \right)^{Ac/4} \right]^m = o(1/n)$$

In particular, it would suit our needs if $A > 6e$ and $Ac/4 \geq 3$. Let us assume $A \geq 16.4$. Under this assumption for small m the above clearly holds. For large m , say greater than $\sqrt{n}$, we want to be able to naively replace the m inside the brackets with n (this only increases the interior value as we are multiplying by m^3 and dividing by m^2) and obtain something smaller than 1, so that we obtain something of the order $(1 - \varepsilon)^{\sqrt{n}} = o(1/n)$. So we want

$$A^2 e^2 \left(\frac{6e}{A} \right)^{Ac/4} < 1.$$

³If R is a binomial random variable $B(n, p)$ then $\mathbb{P}(R \geq k) \leq (npe/k)^k$.

Keeping in mind that the expected size of the graph correlates to A^2c , it is clear that we want to minimise A such that the above holds. Taking $A = 16.4$, $c = 1.85$ suffices. Recalling $b > c$, we know for any $a > b$, whp the graph has at most $A^2an/2$ edges. We therefore take $b = 1.851$ and $a = 1.852$, giving us that

$$\hat{r}(P_n) \leq 250n$$

and completing the proof. $\square$

The constant in the linear bound given by Beck, 900, would go on to be improved several times; it stands at 137 today, as proven by Dudek and Prałat[14].

Bound 18: Linearity of Size-Ramsey number of paths due to Dudek and Prałat 2014.

$$\hat{r}(P_n) < 137n$$

Proof ideas: We will give the necessary condition for a path found by Dudek and Prałat, but omit the actual construction of a host graph.

Lemma 2.3. *Let c be a real number and G a graph of order cn . Suppose G has been coloured in such a manner that there is neither a red copy nor a blue copy of P_n . Then the following two properties hold:*

- *There exist two disjoint sets $R, M \subseteq V(G)$ of size $n(c-1)/2$ each such that there is no red edge between them;*
- *There exist two disjoint sets $R', M' \subseteq V(G)$ of size $n(c-1)/2$ each such that there is no blue edge between them.*

Proof. We will prove the first property; the second holds by a symmetric argument, swapping the colours. We proceed algorithmically: choose a vertex v and initialise $R = V(G) \setminus v$ and $M = \emptyset$ and a third set $P := \{v\}$, which will be a stack.

We now apply a modified version of the DFS algorithm, on the subgraph of red edges. Proceed exactly as before, with the caveat that if the stack P ever empties, then we take a new vertex at random from R and place it in P and reinitialise. As mentioned before, the neighbourhood of M is P and in particular there are no edges from M to R . Halt the algorithm when $M = n(c-1)/2$. At this point, the size of P is at most n by assumption and so the size of R is at least $n(c-1)/2$. We can remove extra vertices if needed to obtain the exact sizes. $\square$

It remains to provide a host graph that does not have one of these two properties whp. Dudek and Prałat remarked that one could take a host graph with the property that, for any two sets S, T of size $n(c-3)/4$, there was at least one edge between these two sets. We omit the proof that such a host graph exists or that it has less than $137n$ edges whp.

Lemma 2.4. *Given a graph G such that for $S, T \subseteq V(G)$, $|S| = |T| = n(c-3)/4$, any colouring of G has a monochromatic copy of P_n .*

Proof. Suppose that there is no monochromatic copy of P_n . Then we know that, by the previous lemma, we can find two disjoint sets $R, M \subseteq V(G)$ of size $n(c-1)/2$ each such that there is no red edge between them, and similarly R', M' for the colour blue. Let $P = (R \cup M)^c$ and $P' = (R' \cup M')^c$.

Now let

$$\begin{aligned} A &:= R \cap R', a := |A|; \\ B &:= R \cap M', b := |B|; \\ C &:= M \cap R', c := |C|; \\ D &:= M \cap M', d := |D|. \end{aligned}$$

Note that these four sets are disjoint. If $(X, Y) \in \{(A, B), (A, C), (B, D), (C, D)\}$ then $x+y = |X \cup Y| \geq n(c-3)/2$. This is as all of these four pairs are of the form $(T \cap U, U \cap V)$ and so $|X \cup Y| = |U \setminus P'|$ (or P , depending on the case) and $|U \setminus P| \geq |U| - |P| = n(c-3)/2$.

Now we observe that either a and d are both greater than or equal to $n(c-3)/4$ or b and c are. The reason is simple: if at least one from each pair is smaller than $n(c-3)/4$, then the two together are not greater than or equal to $n(c-3)/2$, but we just showed this for all four possibilities. However, there are no red or blue edges (and hence no edges) between either of these pairs, and this contradicts our assumption on G . $\square$

Dudek and Pralat took as a host graph $G = \mathcal{G}(An, b/n)$, $A = 7.29$, $b = 5.14$. The curious reader may try and prove that this graph has the properties noted above themselves. $\square$

Beck also conjectured in his paper that the size-Ramsey number of bounded trees was linear. He proved that there were graphs with $D \cdot n \cdot (\log n)^{12}$ edges that contained *all* trees of size n and maximal degree D . His proof of this contained many new ideas that he would go on to repurpose in his proof that there were host graphs inducing all bounded-degree trees, which we will discuss in the next section.

Friedman and Pippenger[25] would go on to prove Beck's conjecture three years later.

Bound 19: Linearity of Size-Ramsey number of bounded-degree trees due to Friedman and Pippenger 1986.

For trees $T_{D,n}$ of maximal degree D and size n ,

$$\hat{r}(T_{D,n}) < C_D n$$

where C_D is a constant depending only on D .

We will not go into the details on the proof but it once again uses expanding graphs. In particular, they showed that graphs with sufficient expander properties (with the adjacent

vertices to a set S being several⁴ times the size of S , rather than as big, as we saw in our proof for the linearizability of $\hat{r}(P_n)$ contained *all* trees of bounded degree and size n . With this result, they proceeded to show that host graphs existed that were linear in size but even if you deleted all but a fixed proportion of their edges, maintained the expander property - implying that for any number of colours, the size-Ramsey number is linear⁵.

Beck had noted in his paper that some form of graph “largeness” control was important; indeed, it is not difficult to see that there are trees with quadratic size-Ramsey number. Consider for instance the bistar, the graph on $2n - 1$ edges consisting of two adjacent vertices of degree n . Taking any host graph that always gives a bistar when coloured in two colours, simply colour edges red if they connect two vertices of degree n and blue otherwise. It is clear that any copy of a bistar must be red, and so we need at least $2n$ vertices of degree n in our host graph, implying at least n^2 edges, which is quadratic in the size of the bistar. Friedman and Pippenger’s constant in their bound would subsequently be improved several times; finally, Dellamonica[9] proved one could calculate a simple function that depended on the bipartition of the tree (and the degrees and sizes of the two sides of the partition) which controlled the size-Ramsey number.

2.3 More general graphs

As we shall see in the next section, it was further proven that the size-Ramsey number of cycles was also linear; this is a simple implication of Haxell, Kohayakawa and Łuczak’s proof[28] that the induced Ramsey and size-Ramsey number of cycles were linear, since induced Ramsey numbers are trivially an upper bound on their general counterparts.

An interesting question to ask then, given the results we have seen so far, is whether all bounded degree graphs have linear size-Ramsey number. Rödl and Szemerédi[36] proved this is not true; they found a sequence of graphs of degree at most 3 such that the size-Ramsey numbers of the graphs formed a superlinear sequence. Kamcev, Liebenau, Wood and Yepremyan[29] proved that graphs with bounded treewidth (where a graph with treewidth k is a subgraph of a chordal⁶ graph that is K_k free, where k is the minimal such number) had size-Ramsey number bounded by their order multiplied by a constant depending on maximum degree and treewidth⁷.

One might also wonder what the best possible bound is when considering graphs with bounded degree greater than 3, since we know it is at least superlinear (degree bounded by

⁴Specifically, the maximum degree of the tree.

⁵Whilst it might seem evident that if the size-Ramsey number is linear for two colours, it would be so for any number of colours via some kind of inductive argument, this is not trivially true. In this case, where you have a sufficient property (being an expander) that you can show is maintained for any subgraph with a sufficient proportion of your edges, it is not difficult to see why induction works, but in other cases this becomes less clear.

⁶A chordal graph has at least one crossing edge in every cycle of length at least 4 has at least one crossing edge. In other words, every cycle is “triangulated”, and the only possible induced cycles are triangles.

⁷The treewidth is a measure of how similar to a tree a graph is. Treewidth 1, for example, is equivalent to being a tree (as otherwise having a cycle forces the presence of a triangle) whereas the higher the treewidth, the more you have cliques and dense structures (for example, the wheel W_n has treewidth 2). Treewidth is an easier notion to use algorithmically than similar measures that also relate to how similar to a tree you are, like k -connectedness, which is why it is easier to work with when proving bounds.

2 of course corresponds to paths or cycles). Of particular interest is the case with bounded degree 3. The current best bound is $O(n^{3/2+o(1)})$, due to Draganić and Petrova[13] in 2022, and the authors remark that it is probably the limit of existing methods.

Finally, for completeness, we include a few select known exact size-Ramsey values without discussing their proof. The constructions for the lower bounds are fun puzzles in and of themselves and the reader is free to think about them. The first concerns forests of stars, and was found by Burr, Erdős, Faudree, Rousseau and Schelp[5].

Result 8: Forests of stars due to Burr and Erdős and Faudree and Rousseau and Schelp.

$$\hat{r}(sK_{1,n}, tK_{1,n}) = (s + t - 1)(n + m - 1).$$

The second, from Faudree and Sheehan[21], is about either having two red edges adjacent to a vertex, or a blue clique.

Result 9: Vee-graph versus clique due to Faudree and Sheehan

$$\hat{r}(K_{1,2}, K_n) = 2(n - 1)^2.$$

The last result was found by the previous two authors working with Rousseau[21] and is one of the rare instances in which a size-Ramsey number is known precisely but not its Ramsey counterpart!

Result 10: Square versus star due to Faudree and Rousseau and Sheehan.

$$\hat{r}(K_{2,2}, K_{1,n}) = 4n.$$

2.4 Conjectures

Whilst we know the size-Ramsey numbers of cycles and bounded-degree trees are all linear for any number of colours (albeit with the constant being a factor of the number of colours k and the degree bound d) it is not known how far we can improve the dependence of the constant on k and d .

3 Induced Ramsey numbers

The chief danger to our philosophy, apart from laziness and woolliness, is scholasticism, which is treating what is vague as if it were precise...

Frank Plumpton Ramsey

The induced Ramsey numbers are much trickier to work with as there is the additional constraint that no additional edges should be present between the vertices of the subgraph. Of course, induced and general Ramsey numbers overlap for the complete case, so we will only consider other graphs in this section. Even selecting the host graph is nontrivial in the induced case - if you want to induce any non-complete graph, you would not be able to do so by using any complete graph as a host graph. If all the edges of the complete graph were red, then every induced red subgraph would be complete and every induced blue subgraph would be empty. It is necessary to remove some edges from the host graph in order to ensure that you do not run into issues. It is clear that induced Ramsey numbers must be bounded below by their general Ramsey counterparts.

The existence of induced Ramsey numbers was itself a nontrivial question and not answered until Deuber[10]; Erdős, Hajnal and Pósa[17], and Rödl[35] all independently proved that valid host graphs always existed in the 1970s. The proof of Erdős, Hajnal and Pósa (and indeed, most proofs for induced Ramsey numbers) involves an inductive embedding of the desired graph into one of the two colours, under the assumption that the desired graph does not exist in the other colour. The host graph is usually asked to have desirable properties in terms of the neighbourhood of vertices (in particular, given any set of vertices S some bounded size, for any subset of S there are sufficiently many vertices connected to only the elements of S and none other). The explicit existence of a graph with such properties can be done in many ways, as we shall see. Beck[3], for example, proved that random graphs would satisfy his requirements with high probability.

A key result used in induced proofs is Szemerédi's regularity lemma. This beautiful result states informally that any large enough graph can be partitioned into at most a constant number of parts, where the edges between parts behave almost randomly. In particular, the edge density between subsets of the different parts almost matches the edge density between the parts themselves.

For this section we will say, for two graphs G and H , that G is H -*inducing* if any colouring of G in two colours yields an induced monochromatic copy of H .

3.1 Cycles and paths

Graham and Rödl[26] were the first to conjecture that a linear upper bound existed for the induced and size-induced number of a path or cycle in any number of colours, following Beck's proof for the linearity of the size-Ramsey number of these graphs that we saw in the previous section. This would be proven by Haxell, Kohayakawa and Łuczak[28] in 1995.

Bound 20: Induced Ramsey number of Cycles due to Haxell et al.

$$\hat{r}_{ind}(C_n) < Cn.$$

Proof ideas: We will only outline the motivation for choice of host graph and main implications of this choice here as it is very long. The host graph is chosen to be a binomial random graph, with in particular a sparse number of expected edges. For some variable N and constant D , take the random graph $\mathcal{G}(N, p)$, $p := D/N$ where each edge occurs independently with probability p . We then delete from this graph all vertices of large degree and edges in short cycles in order to obtain a secondary graph G ; if N is picked so that the size of G , $|V(G)| = n$ satisfies the desired bound then G will be our host graph.

Colour the edges in k colours. Using a variant of Szemerédi's regularity lemma, one can then guarantee that there is some colour i with some desired properties, which are as follows (we simply state the properties, but do not provide any form of proof). Let J be the subgraph of edges in the colour i .

Firstly, we can guarantee the presence of a short induced cycle in J , with no edges of any other colour present either, whose size is of the order $\log n$. We then show that this cycle can be augmented whilst preserving the condition that no edges of any other colour are present - whilst this is a stronger inductive step than needed, the more precise conditions ensure that we can indeed continue to augment the size. In particular, we augment the cycle by length 1 by replacing segments with other segments that are longer by 1.

Notably, we are able to successfully do this because we trimmed our original graph to remove high degrees and short cycles, guaranteeing that longer cycles will not have internal crossing edges of any colour. It can also be shown that if the augmenting step fails then it implies the existence of a subgraph which is relatively dense in the number of edges, which is a contradiction with the removal of all vertices of high degree initially. This construction also implies that the induced Ramsey number for paths is linear. $\square$

One would expect that a simpler proof is possible for paths. A simple way to find an induced path is the diameter of a graph. By definition, the path giving the distance between any two vertices must be the induced subgraph corresponding to the vertices on the path, as if there were any other edge then the distance between the vertices could be reduced. How, then, do we pick a host graph such that one of the two colours will have large diameter? And how to find the diameter, or at evaluate the graph in some manner to understand the distances between vertices?

It turns out that an appealing way to approach the problem is via a classical computer science technique for exploring graphs, found by Draganić, Glock and Krivelevich[12].

Bound 21: Induced Ramsey number of Paths due to Draganić and Glock and Krivelevich.

$$\hat{r}_{ind}(P_n) < 5 \cdot 10^7 n.$$

Proof ideas: Similarly to before, we want to apply the DFS algorithm. However, this time, we want to ensure that the stack is an induced path. By adding the simple step where we check if new members of stack have additional neighbours in the stack (apart from the previous newest member), we can ensure that the stack is always an induced path. Rather than having a single set of completely explored vertices, we now have two sets of explored vertices - one for the completely explored vertices and one for those we remove from the stack for the secondary reason that they did not induce a path.

- Our input will be a graph $G = (E, V)$ and a vertex $v \in V$. This time we will work with four sets - a set of “unexplored” vertices R , a set of vertices S which have been explored along with all their neighbours, a set of vertices U that were discarded because they did not induce a path, and a stack T of vertices, which is a data structure where we add and remove from the top of the stack - in other words, the first vertex to leave is the most recent arrival. Initially, v is in T and every other vertex is in S .
- At every stage, we look at the top of the stack T and if the vertex v there has at least one neighbour w in R , we add one at random to the stack. Otherwise, we remove the vertex at the top of the stack T and add it to S . However, if w has additional neighbours in T , we discard it to U .

The stack, at any given moment, is an induced path. We know for sure there are no edges between S and R , and that every time we add a vertex to the stack, the size of $T \cup S \cup U$ increases by 1 (and never decreases), with the number of edges between them being at least $2|U|$ since every vertex in U has at least two neighbours in R when it first arrives, and those neighbours cannot come to U at a later point since being sent to U can only happen the first time you enter the stack, and so they will either be in U or S .

Now, we remark that letting the algorithm run on suitable graphs will always give sufficiently long induced paths. More specifically, let u, s, t be constants and take a graph G of order at least $u + s + t$ where induced graphs of order less than $u + s + t$ have less than $2u$ edges. Also assume that we have a subgraph H where, if $W \subset H$ and $|W| = s$, $|N_H(W)| \geq t + u$. We claim that H has a path of length at least t that is induced in G .

Run the algorithm but terminate when S has size s or U has size u for the first time. Suppose at this point T has size less than t . If we terminated because U became full, then we have at least $2u$ edges in $U \cup S \cup T$, contradicting our first assumption on the graph. If we terminated because S became full, the neighbourhood of S is a subset of $U \cup T$ and so smaller than $u + t$, contradicting our second assumption. Note that we must terminate, since vertices will all eventually end up in S or U .

How would we quantify the desired properties in terms of events that are simple to assess probabilistically? One way is to show that the host graph has high minimum degree, low edge density in all small sets and a low number of crossing edges between all pairs of small sets.

Suppose we now take the random graph $\mathcal{G}(n, C/n)$ where C is a constant we will determine later. Let us colour the edges arbitrarily and take the subgraph H of the more common

colour. Firstly, we can trim all the isolated vertices of H (by abuse of notation, let us henceforth call this new graph H again) and we know that this removes $\Sigma(n)$ vertices whp so that we are left with less than $(1 - \epsilon)n$ but more than $(1 - \delta)n$ vertices for some $\delta > \epsilon > 0$. Note that these values δ, ϵ are not a function of n but universal constants.

Furthermore, the graph G has whp at least $(1 - o(1))Cn/2$ edges, implying the graph H has whp at least $(1 - o(1))Cn/4$ edges. If we now continually remove vertices of degree at most $Cn/4$ until there are none left then we will be left with a nonempty graph, since every time we remove a vertex we remove at most $Cn/4$ edges and H has more than $Cn/4 \cdot |H| = Cn/4(1 - \epsilon)n$ edges. This nonempty graph has high minimum degree, which was one of our initial goals.

Now let us ensure that small sets have low edge density. Let $c < 1$ be another constant, again to be determined later. For what values $d > 1$ can we say that all sets S of size at most cn span less than $d|S|$ edges? Let us apply a simple union bound on the set of bad events:

$$\begin{aligned} \sum_{i=1}^{cn} \binom{n}{i} \binom{\binom{i}{2}}{di} \cdot (C/n)^{di} &\leq \sum_{i=1}^{cn} \left(\frac{en}{i}\right)^i \left(\frac{eiC}{2dn}\right)^{di} = \sum_{i=1}^{cn} \left[\left(\frac{en}{i}\right) \cdot \left(\frac{eiC}{2dn}\right)^d \right]^i \\ &\leq \sum_{i=1}^{cn} \left[\left(\frac{C}{2d}\right)^d e^{1+d} \cdot \left(\frac{i}{n}\right)^{d-1} \right]^i. \end{aligned}$$

Note that in the final expression, most of the terms are constant save the $(\frac{i}{n})^{d-1}$. It would be nice to replace the i with cn , and if the resulting expression inside the brackets evaluates to something smaller than 1 then we are left with $\sum_{i=1}^{cn} (1 - \epsilon)^i$. Unfortunately, the individual summands do not become order $o(1/n)$ until i is sufficiently large. However, we can avoid this issue by dominating the first $\sqrt{n}$ terms of the sum; notably, the terms would be dominated by $\sum_{i=1}^{\sqrt{n}} n^{-(d-0.5)/2}$ which tends to 0 as $n \rightarrow \infty$ for $d < 2.5$. It is therefore additionally only necessary that $(\frac{Cec}{2d})^d e/c < 1$ in order for the bad events to not happen whp.

Finally, we want to bound the number of crossing edges between small sets. Let $a < c$ and $b < 1$ be two more constants to be determined later and let us again calculate, via the union bound, the valid f such that for all disjoint sets S, T with $|S| = an$ and $t \leq bn$, we can say there are at most $f|S|$ edges between the two whp. Note that it suffices to consider $t = bn$. The set of bad events has probability:

$$\begin{aligned} \binom{n}{an} \binom{n}{bn} \binom{abn^2}{fan} (C/n)^{fan} &\leq \left(\frac{ne}{an}\right)^{an} \left(\frac{ne}{bn}\right)^{bn} \left(\frac{abn^2e}{fan}\right)^{fan} \left(\frac{C}{n}\right)^{fan} \\ &\leq \left((e/a)^a (e/b)^b (beC/f)^{fa} \right)^n \end{aligned}$$

Again, if the constant term is less than 1 then we can assume bad events do not happen.

Now, suppose that we picked a, b, c, C, d and f judiciously that satisfied the above equations. It would then follow that if a set of vertices S of size an had a neighbourhood T of size less than bn , then we would know the following.

- The number of edges from S to T is at least the minimum degree of S multiplied by the size of S , minus all internal edges (counted twice).
- This is at least $(C/4 + 1)an - 2dan$ (recall that we took $a < c$), allowing us to use the low edge density property.
- If we now have that $(C/4 + 1)an - 2dan > fan$, this is a contradiction with the low crossing edges property.

We would therefore pick our variables in order to satisfy the above conditions, along with the new inequality $(C/4 + 1)an - 2dan > fan$. We omit the precise calculations, but this is indeed possible. To finish the proof, note that running the modified DFS algorithm on this graph gives a path of length n , and that whp the graph has at most a linear number of edges in n . $\square$

Note that in particular, for the multicoloured case, the constants implied by the proof of Haxell, Kohayakawa and Luczak are huge in terms of their dependence on k . They have since significantly been improved; the current best bounds for paths are those in the paper of Draganić, Glock and Krivelevich. The best bounds for cycles were found by Bradač, Draganić and Sudakov[4] recently and show that for n even, C_k can be bounded by a polynomial in k and by $k^{\tilde{O}(k)}$ for the odd case.

3.2 Trees and bounded-degree graphs

Beck[3] was the first to prove a result on induced size-Ramsey numbers for trees.

Bound 22: Induced Ramsey number of Trees

There exists a graph G with less than $n^3(\log n)^4$ edges such that G is T_n -inducing for *every* tree T_n of order n .

Proof. Like with previous proofs, we will try and identify specific graph properties that are desirable and imply the existence of an induced tree.

Here, we will seek a different kind of expander property. Our aim will be to embed the tree T_n into the larger monochromatic subgraph of G , inductively. In order to do so, it will be necessary that at each step, if we want to attach a leaf to our existing tree T_k at some vertex v , there is a vertex u adjacent to v but not T_k .

We could simply ask for the condition that for any vertex $v \in V(G)$ and set $S \subset V(G)$, $|S| < n$ that $\Gamma_G(v) \setminus \Gamma_G(S) \neq \emptyset$. However, it is hard to see how we can assure that this property holds in the host graph without a vast number of edges.

We will therefore reuse an old trick: we will prove a lemma similar to lemma 2.2 from the previous section.

Lemma 3.1. *Consider a graph G with average degree d . Then G has an induced subgraph $G[H]$ for some $H \subseteq V(G)$ such that the minimum degree of H is at least $d/2$.*

Proof. Let X be a minimal subset of $V(G)$, under set inclusion, for the property that $|G[X]| > d \cdot |X|/2$. We claim X is the desired H . Assume this is not the case; then there exists $v \in X$ such that the degree of v in X , $d_X(v) < d/2$. But if this is the case, $X \setminus v$ has the property that $|G[X \setminus v]| > d \cdot |X \setminus v|/2$ and is a subset of X , contradicting its minimality. $\square$

Via this lemma, we see that we can be sure that one of the two colours has an induced subgraph with high minimal degree, which we might then use to find an embedding of T_n . It remains to adapt our property to take into account this.

Lemma 3.2. *Consider a graph G with average degree d with the following property: for any vertex $v \in V(G)$ and set $S \subset V(G)$, $|S| < n$, we have*

$$|\Gamma_G(v) \cap \Gamma_G(S)| \leq d/4.$$

Then G is T_n -inducing.

Proof. Colour G in two colours and let H be the monochromatic subgraph of G with more edges. It follows that the average degree of H is at least $d/2$, and by the above lemma, we can find an induced subgraph $F[H]$ such that the minimum degree of $F[H]$ is $d/4$.

Now we simply successively embed the tree T_n into $F[H]$. Note that we can write a sequence of trees $T_1, T_2, T_3, \dots, T_n$ where T_1 is a single vertex and T_i is $T_{i-1} \cup (u, v)$, $u \in V(T_{i-1})$, $v \notin V(T_{i-1})$ - the previous tree with a new leaf.

Suppose T_k can be embedded into $F[H]$ and that the new leaf is (u, v) . We know that by the assumed property on G ,

$$|\Gamma_G(u) \cap \Gamma_G(V(T_k) \setminus u)| \leq d/4$$

implying, as $d_F(u) \geq d/4$, that there is a vertex $v \in \Gamma_F(u) \setminus \Gamma_G(V(T_k) \setminus u)$ adjacent to u and present in F but not adjacent to T_k . We can embed the new leaf using the edge (u, v) . $\square$

It remains to be shown that a host graph exists with this property. Consider a binomial random graph $G = \mathcal{G}(n^2(\log n)^2, 1/(18n))$. Let $M := n^2(\log n)^2$, $p := 1/(Cn)$ for some constant C we will deduce later.

Firstly, we see that we can assume whp the maximum degree of G is less than $3Mp$. We can show this via a union bound on the bad events - let A_x be the event that vertex x has degree at least $3Mp$; by the same Chernoff bound used in the previous section we know

$$\mathbb{P}(A_n) \leq (Mpe/3Mp)^{3Mp} = (1 - \epsilon)^{O(M)} = o(1/M)$$

and in particular the union bound over all vertices gives the probability the maximum degree of G is greater than $3Mp$ is $o(1)$. Moreover, the expected number of edges is

asymptotically $M^2p/2$, so we can say by our usual concentration inequalities that whp $(1-d)M^2p/2 < |G| < (1+d)M^2p/2$ for any choice of d . Notably, whp the average degree of G is at least $Mp/2$, and whp it has at most M^2p edges.

Now we will show, under these two assumptions, that the probability our random graph does not have the property above is less than 1. Clearly, we can assume $|S| = n - 1$. A bad event is therefore a choice of $n - 1$ vertices for S and another vertex for v such that their shared neighbourhood contains at least $Mp/2$ vertices. In other words, there are $Mp/2$ vertices connected to both S and v . Consider $N_G(v)$; we know $|N_G(v)| < 3Mp$. The probability that at least $Mp/2$ of these vertices are connected to S is

$$\mathbb{P}(|G[S, N_G(v)]| > Mp/2) \leq \mathbb{P}(B(3Mp(n-1), p) > Mp/2)$$

using the standard notation $G[X, Y]$ for the subgraph of edges with one endpoint in X and the other in Y . Again, we apply Chernoff, and see that

$$\begin{aligned} \mathbb{P}(B(3Mp(n-1), p) > Mp/2) &\leq (6Mp^2(n-1)e/Mp)^{Mp/2} \\ &\leq (6ep(n-1))^{Mp/2} \\ &= (1-c)^{Mp/2} = o(1/(M^n)) \end{aligned}$$

for some constant $0 < c < 1$, if we take $C > 6e$ - for example, $C = 18$. But then it is easy to see that the total probability of bad events is less than $M^n \cdot o(1/(M^n)) = o(1)$, as there are at most M^n ways to select S and v . It follows that G has the desired property and size at most

$$M^2p < n^3(\log n)^4.$$

□

Fox and Sudakov[23] showed that there were trees whose induced Ramsey number was superlinear in the number of vertices. They used the tree consisting of a star attached to a path and had the novel idea of bounding it by the weak induced Ramsey number, wherein the induced graph is allowed to include edges of the other colour. Specifically, they remarked that $r_{ind}(T) > r_{weak\ ind}(P_{n/2}, K_{1, n/2})$ where T was the graph obtained by attaching the center of the star to the path. They then used the regularity lemma to show that sufficiently large graphs could be coloured in such a manner that the blue graph had bounded maximum degree (with the bound being a linear function of n) and the red induced subgraphs either had large numbers of edges or independent sets. This prevents blue induced stars or red induced paths occurring with high probability.

For general graphs with bounded degree, Łuczak and Rödl[32] proved the induced Ramsey number was at most polynomial in the order (with the exponent being a function of the maximal degree), proving a conjecture of Trotter.

3.3 Conjectures

In their 2023 paper, Bradač, Draganić and Sudakov conjecture that the bound for odd cycles could be reduced to $n \cdot O(e^k)$. They also remark that their proof (which uses a host

graph constructed by taking a hypergraph and placing copies of a smaller graph with desirable properties, namely that it induces pentagons when coloured with k colours) could be modified to prove this, by replacing the small graph with one that has $e^{O(k)}$ edges and induces a copy of a monochromatic odd cycle (larger than a triangle!). Indeed, the proving this bound for fixed odd cycles is also an open question, but this version is weaker and therefore more approachable.

It is conjectured that for bounded degree trees, the induced size-Ramsey number is $n^{1+o(1)}$; in other words, it grows faster than any linear sequence but slower than any superlinear power of n . The lower bound has already been proven as seen before but the upper bound remains open.

References

- [1] M. Ajtai, J. Komlos, and E. Szemerédi. “A Dense Infinite Sidon Sequence”. In: *European Journal of Combinatorics - EJC* 6 (Mar. 1981). DOI: 10.1016/S0195-6698(81)80014-5.
- [2] J. Beck. “On size Ramsey number of paths, trees, and circuits. I”. In: *Journal of Graph Theory* 7.1 (1983), pp. 115–129.
- [3] József Beck. “On Size Ramsey Number of Paths, Trees and Circuits. II”. In: *Mathematics of Ramsey Theory*. Ed. by Jaroslav Nešetřil and Vojtěch Rödl. Berlin, Heidelberg: Springer Berlin Heidelberg, 1990, pp. 34–45.
- [4] D. Bradač, N. Draganić, and B. Sudakov. *Effective bounds for induced size-Ramsey numbers of cycles*. 2023. arXiv: 2301.10160 [math.CO].
- [5] S.A. Burr et al. “Ramsey-minimal graphs for multiple copies”. In: *Indagationes Mathematicae (Proceedings)* 81.1 (1978), pp. 187–195.
- [6] M. Campos et al. *An exponential improvement for diagonal Ramsey*. 2023. arXiv: 2303.09521 [math.CO].
- [7] V. Chvátal. “Tree-complete graph ramsey numbers”. In: *Journal of Graph Theory* 1.1 (1977), pp. 93–93.
- [8] D. Conlon. “A new upper bound for diagonal Ramsey numbers”. In: *Annals of Mathematics* 170.2 (2009), pp. 941–960.
- [9] D. Dellamonica Jr. “The size-Ramsey number of trees”. In: *Random Structures & Algorithms* 40.1 (2012), pp. 49–73.
- [10] W. Deuber. “A generalization of Ramsey’s theorem”. In: *Colloquia Mathematica Societatis János Bolyai* 10 (1975), pp. 323–332.
- [11] G. A. Dirac. “A Property of 4-Chromatic Graphs and some Remarks on Critical Graphs”. In: *Journal of the London Mathematical Society* s1-27 (1952), pp. 85–92.
- [12] N. Draganić, S. Glock, and M. Krivelevich. “Short proofs for long induced paths”. In: *Combinatorics, Probability and Computing* 38 (2022), pp. 870–878.
- [13] N. Draganić and K. Petrova. *Size-Ramsey numbers of graphs with maximum degree three*. 2022. arXiv: 2207.05048 [math.CO].

- [14] A. Dudek and P. Pralat. “An Alternative Proof of the Linearity of the Size-Ramsey Number of Paths”. In: *Combinatorics, Probability and Computing* 24.3 (2014), pp. 551–555.
- [15] P. Erdős. “Some remarks on the theory of graphs”. In: *Bulletin of the American Mathematical Society* 53.4 (1947), pp. 292–294.
- [16] P. Erdős and R.L. Graham. “On Partition Theorems for Finite Graphs”. In: *Colloquia Mathematica Societatis János Bolyai* 10 (1973), pp. 515–527.
- [17] P. Erdős, A. Hajnal, and L. Pósa. “Strong embeddings of graphs into colored graphs”. In: *Infinite and Finite Sets* 10 (1975).
- [18] P. Erdős and G. Szekeres. “A combinatorial problem in geometry”. In: *Compositio Mathematica* 2 (1935), pp. 463–470.
- [19] P. Erdős et al. “On Partition Theorems for Finite Graphs”. In: *Periodica Mathematica Hungarica* 9 (1978), pp. 145–161.
- [20] R.J. Faudree and R.H. Schelp. “All Ramsey numbers for cycles in graphs”. In: *Discrete Mathematics* 8.4 (1974), pp. 313–329.
- [21] R.J. Faudree and J. Sheehan. “Size Ramsey numbers involving stars”. In: *Discrete Mathematics* 46.2 (1983), pp. 151–157.
- [22] R.J. Faudree et al. “Path-cycle Ramsey numbers”. In: *Discrete Mathematics* 10.2 (1974), pp. 269–277.
- [23] Jacob Fox and Benny Sudakov. *Induced Ramsey-type theorems*. 2007. arXiv: 0706.4112 [math.CO].
- [24] P. Frankl and R.M. Wilson. “Intersection theorems with geometric consequences”. In: *Combinatorica* 1 (1981), pp. 357–368.
- [25] J. Friedman and N. Pippenger. “Expanding graphs contain all small trees”. In: *Combinatorica* 7 (1987), pp. 71–76.
- [26] R.L. Graham and V. Rödl. “Numbers in Ramsey theory”. In: *Unknown* (Jan. 1987), pp. 111–153. URL: https://mathweb.ucsd.edu/~fan/ron/papers/87_04_ramsey_numbers.pdf.
- [27] J.E. Graver and J. Yackel. “An upper bound for Ramsey numbers”. In: *Bulletin of the American Mathematical Society* 72.6 (1966), pp. 1076–1079.
- [28] P. E. Haxell, Y. Kohayakawa, and T. Łuczak. “The Induced Size-Ramsey Number of Cycles”. In: *Combinatorics, Probability and Computing* 4.3 (1995), pp. 217–239.
- [29] N. Kamcev et al. “The Size Ramsey Number of Graphs with Bounded Treewidth”. In: *SIAM Journal on Discrete Mathematics* 35.1 (2021), pp. 281–293.
- [30] J.H. Kim. “The Ramsey number $R(3, t)$ has order of magnitude $t^2/\log t$ ”. In: *Random Structures & Algorithms* 7.3 (1995), pp. 173–207.
- [31] Gerencsér L. and A. Gyárfás. “Tree-complete graph ramsey numbers”. In: *Journal of Graph Theory* 1.1 (1977), pp. 93–93.
- [32] T. Łuczak and V. Rödl. “On Induced Ramsey Numbers for Graphs with Bounded Maximum Degree”. In: *Journal of Combinatorial Theory, Series B* 66 (1996), pp. 324–333.

- [33] J. Nešetřil and V. Rödl. “The Ramsey property for graphs with forbidden complete subgraphs”. In: *Journal of Combinatorial Theory, Series B* 20.3 (1976), pp. 243–249.
- [34] F.P. Ramsey. “On a Problem of Formal Logic”. In: *Proceedings of The London Mathematical Society* (1935), pp. 264–286.
- [35] V. Rödl. “The dimension of a graph and generalized Ramsey theorems [EN, original CZ]”. In: *Master’s Thesis, Charles University* (1973).
- [36] V. Rödl and E. Szemerédi. “On Size Ramsey Numbers of Graphs with Bounded Degree”. In: *Combinatorica* 20 (2000), pp. 257–262.
- [37] A. Sah. “Diagonal Ramsey via effective quasirandomness”. In: *Duke Mathematical Journal* 172.3 (2023), pp. 545–567.
- [38] J. Spencer. “Ramsey’s theorem — A new lower bound”. In: *Journal of Combinatorial Theory, Series A* 18.1 (1975), pp. 108–115. ISSN: 0097-3165.
- [39] A. Thomason. “An upper bound for some ramsey numbers”. In: *Journal of Graph Theory* 12.4 (1988), pp. 509–517.



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every semester paper, Bachelor's thesis, Master's thesis and any other degree paper undertaken during the course of studies, including the respective electronic versions.

Lecturers may also require a declaration of originality for other written papers compiled for their courses.

I hereby confirm that I am the sole author of the written work here enclosed and that I have compiled it in my own words. Parts excepted are corrections of form and content by the supervisor.

Title of work (in block letters):

RAMSEY, SIZE-RAMSEY AND INDUCED SIZE-RAMSEY
NUMBERS

Authored by (in block letters):

For papers written by groups the names of all authors are required.

Name(s):

~~Patil~~ PATIL

First name(s):

TANISH

With my signature I confirm that

- I have committed none of the forms of plagiarism described in the 'Citation etiquette' information sheet.
- I have documented all methods, data and processes truthfully.
- I have not manipulated any data.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for plagiarism.

Place, date

Geneva, 31.08.2023

Signature(s)

For papers written by groups the names of all authors are required. Their signatures collectively guarantee the entire content of the written paper.